NYU | SCHOOL OF GLOBAL PUBLIC HEALTH

Lecture 3 – Measures of Association

Farzana Kapadia, PhD MPH
Professor of Epidemiology

1

Lecture objectives

- To review measures of associations
- To review measures of effect
- To discuss choice of measures in public health and epidemiologic studies
- To review appropriate risk communication approaches
- To discuss relations between measures

2

The 2x2 table

	Outcome	No outcome	Total	Risk	
Exposed	60	140	200	**0.30**	
Unexposed	15	85	100	**0.15**	
Total					

- Reminders:
  1. Row/columns labels can be switched!
  2. We can only calculate risks if we assume: (a) a fixed time-period of observation and (b) everyone followed up without losses for that time-period
  3. Risk Ratio = 2
  4. Risk Difference = 0.15 or 15 per 100

3

Identifying comparison groups to contrast

- To estimate a contrast, must have at least two groups.
  - What do you do when you have more than 2 groups?
  - What about levels of exposure for a continuous measure?
  - What about nominal factors (i.e. biological sex, gender identity, etc.)
    - *This becomes trickier when you start thinking of causal contrasts.*
- Contrasts requires assignment to group, which requires measurement of group membership
  - What will happen if the exposure is measured poorly?
    - *For the 'exposed' group?*
    - *For the 'unexposed' group?*
- *How do these measurement issues impact how we create our research questions?*

4

Example: identifying an appropriate comparison group in an RCT

- Background: Prior literature suggests that women who use injectable hormonal contraception DMPA (Depo-Provera) are at increased vulnerability for HIV acquisition
- An RCT is proposed
  - Study population: ??
  - Treatment arm: ??
  - Appropriate control arm: ??
- Choice of comparison group poses a slightly different scientific question
  - is each question more biological, or does it involve behavioral aspects?
  - how would an analysis treat unintended pregnancy in each arm?
  - how would answers to each question be applied in the real world?

5

Measures of Association I: Risk Contrasts

- Risk over N years for outcome Y when X=1 → $P(Y=1|X=1) = P(Y|X=1)$
- Risk over N years for outcome Y when X=0 is → $P(Y=1|X=0) = P(Y|X=0)$
  - **RISK DIFFERENCE** → $P(Y|X=1) - P(Y|X=0)$
    - the difference in risks comparing exposed and unexposed individuals
    - Range: -1 to 1
    - Null value: 0
  - **RISK RATIO** → $P(Y|X=1) / P(Y|X=0)$
    - the ratio of risks comparing exposed and unexposed individuals
    - Range: 0 to ∞ (or undefined, depending)
    - Null value: 1

6

Measures of Association II: Rate Contrasts

- Rate of outcome Y when X=1 → IR(Y=1|X=1) = IR(Y|X=1)
- Rate of outcome Y when X=0 → IR(Y=1|X=0) = IR(Y|X=0)
  - **RATE DIFFERENCE** → IR (Y|X=1) - IR(Y|X=0)
    - the difference in rates comparing exposed and unexposed individuals
    - Range: -∞ to ∞
    - Null value: 0
  - **RATE RATIO** → IR(Y|X=1) / IR(Y|X=0)
    - the ratio of rates comparing exposed and unexposed individuals
    - Range: 0 to ∞
    - Null value: 1

7

Measures of Association III: Odds Contrasts

- Odds over N years for outcome Y when X=1 → P(Y=1|X=1) / P(Y=0|X=1) = Odds(Y|X=1)
- Odds over N years for outcome Y when X=0 → P(Y=1|X=0) / P(Y=0|X=0) = Odds(Y|X=0)
  - **Odds difference** → **Doesn't exist! Can't do it! Doesn't make sense!**
  - **ODDS RATIO** → Odds (Y|X=1) / Odds (Y|X=0)
    - the ratio of odds comparing exposed and unexposed individuals
    - Range: 0 to ∞
    - Null value: 1

8

Differences, ratios, and public health

- Ratio measures are independent of absolute risks
- An example:
  - Say, risk ratio for lung cancer comparing smokers to non-smokers is 5.
  - How many lung cancer cases will we see among 1000 smokers? 1000 nonsmokers?
    - You don't know from the risk ratio alone

9

Differences, ratios, and public health

- It is generally held that if you're planning a public health intervention, what you care about is the number of lives saved (or cases of disease presented), and the number of people on whom you need to intervene for those purposes.
- Compare 20% vs. 10%, and 4% vs. 2%
  - Same risk ratios (RRs: 2.00 and 2.00)
  - Very different risk differences (RDs: 10% and 2%)
  - Which is "right"? Which is more useful? For what purpose?

10

Risk ratio vs. odds ratio

- Compare 20% vs. 10%, and 4% vs. 2%
  - Same risk ratios (RRs: 2.00 vs. 2.00)
  - Different odds ratios (ORs: 2.25 vs. 2.04)
- Which is “right”? Which is more useful? For what purpose?
  - Remember, OR will always approximate RR when risk is low, or RR is null.
  - When risks are large and non-null, the OR will overestimate RR.

11

Reminder: why do odds overstate risks?

- Recall, a risk is P where $0 \le P \le 1$. You could also state P as P/(1)
- Odds is P/(1-P).
  - When P=0, risk = odds = 0
  - When P>0, that is $0 < P \le 1$, (1-P) is always < 1
  - So except when both are zero, odds always has a smaller denominator than risk – and so is always larger than risk. So, odds generally overstate risks.
  - The bigger the risk, the smaller the denominator of the odds → the bigger the risk, the odds will overestimate the risk by a greater degree.

12

More on risk ratios vs. odds ratios

- If risk ratio is 2.0, what is the odds ratio?
  - Can't be answered without knowledge of baseline risks.
  - No general correction factor to convert RR to OR and back again.

- An advantage of the odds ratio?
  - Risks are limited to the 0 to 1 range, while odds are in the 0 to ∞.
  - As a result, the range of the risk ratio is limited by the baseline risk.
  - If baseline risk is 50%, what is the greatest possible risk ratio? The greatest possible odds ratio?

13

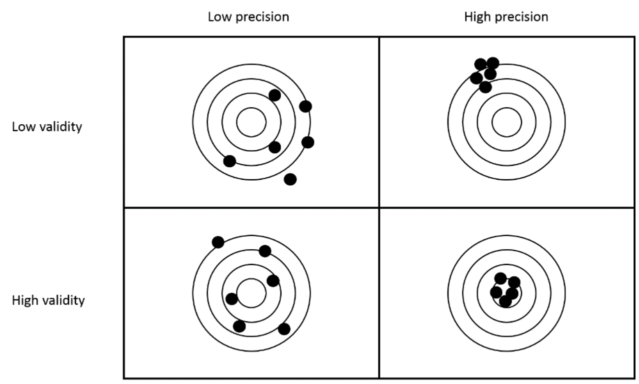
Introduction to precision and confidence intervals

- So far, we have only discussed estimating a contrast (and previously, a measure of disease)
  - But of course, our measures are estimated with error
  - Estimating precision is as important as estimating the association itself
    - Quite possibly, more important

14

Bias and variance

- Bias is a systematic error
  - due to systematic errors, such as “confounding” or “selection bias”
  - systematic errors are those which <u>do not get smaller</u> as sample size gets larger
- Variance is random error
  - sampling error
  - random errors <u>do get smaller</u> as sample size gets larger



15

Understanding variance

- Two main ways:
  - P-values
  - 95% confidence intervals
- Neither is defined precisely the way you want it to be.

We are NOT going to address calculations of these measures in this class; I am leaving that for your biostatistics coursework and there are various places online if you want them. Here, I will address only interpretations of these statistics.

16

P-values

- P-values are used as tests of a hypothesis
  - Usually, of a "null hypothesis", that there is no difference in risk (e.g. that RD = 0, or RR = OR = 1) of outcome comparing exposed and unexposed people
- P is the probability that given the null hypothesis is true, we would see results as extreme or more extreme than the observed results by chance alone.
  - We typically set a p-value in advance (very often, arbitrarily, 0.05) and if the observed p-value is <= 0.05, we reject the null hypothesis
  - *What are some problems with using p-values?*

17

95% confidence intervals

- **If study is repeated infinite times in the same population, then 95% of the confidence intervals for the OR would include the population OR.**
- **The data are consistent with the range of estimates within the confidence interval.**

- A range of **plausible values** for a population parameter which depends on the **point estimate**, **its variability**, and the sample size.

RD, RR, OR

Standard Error:
- **A calculation based on your data**
- **A range of plausible values for a population parameter which depends on the point estimate, its variability, and the sample size**

- It is NOT defined as "we are 95% sure that the true parameter is in this range"
  - Not quite. For that, you could rely on a Bayesian credible interval

18

How to communicate risk and rate ratios

- Be super, super obvious. Way more obvious than you think is necessary.
- For example:
  - The 3-year risk of death among the exposed was 1.5 times the risk among the unexposed.
  - The 5-year risk of death among the exposed was 10 percentage points higher than the risk among the unexposed.
  - The one-month risk of recurrent heart attack among the exposed was five times the one-month risk among the unexposed.
  - The incidence rate of serious injury among the exposed was 1.5 times that among the unexposed.
- There are numerous ways to be right (and wrong) here. Be clear above all, for your audience.
  - Corollary: ALWAYS know who your audience is!!

19

Common issues in communicating risks

- Failing to state the time-period associated with a risk.
  - This often makes risks uninterpretable: everyone dies, so the risk (in exposed and unexposed) of death without a specific time period is 100%.
- Failing to include a comparison group! It's a contrast! This is important!
- Saying "…risk of death among the exposed was 10% higher than among the unexposed."
  - "10% higher" could mean risk difference of 10%, but that could also mean risk ratio of 1.10.
    - Use "10 percentage points" or language about "on the absolute scale" to indicate differences
    - use "1.1 times the risk" or similar language to indicates ratios.

20

Contrast are averages

- Critical reminders:
  - All contrast measures are averages.
  - They cannot be assumed to apply directly to any individual.
  - In fact, sometimes they apply to no individuals.
- For example, say that a particular drug is associated with 10 points lower LDL cholesterol in people with XX chromosomes, and with 10 points higher LDL cholesterol in people with any other set of sex chromosomes.
  - Looking at the association of this drug and LDL change in a population of half XX, half XY + other, would yield an average association of 0 points LDL change.
  - Yet there is no single individual in the studied population (in any population) in whom the drug would lower cholesterol by zero points.

21